

MSC2050 Discrete Mathematics, Presentation 7

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Solving Congruences

(Paragraph 4.4)

Linear congruences

Let a and b be integers, and x be a variable. A congruence of the form

$$ax \equiv b \pmod{m},$$

where m is a positive integer, is called a **linear congruence**.

An integer $\bar{a}$ such that

$$\bar{a}a \equiv 1 \pmod{m},$$

if it exists, is said to be an **inverse of a modulo m** .

Use inspection when m is small

Find inverse of 3 modulo 7.

We need to find j such that $j \cdot 3 \equiv 1 \pmod{7}$

or j such that $7 \mid j \cdot 3 - 1$.

$j = -2$ or $j = 5$.

Existence and uniqueness of inverse

Theorem

If a and m are relatively prime integers and $m > 1$, then an inverse of a modulo m **exists**.

Furthermore, this inverse is **unique** modulo m (any other inverse is congruent to this one modulo m).

Proof: Because $\gcd(a, m) = 1$, there are integers s and t such that $sa + tm = 1$. This implies that $sa + tm \equiv 1 \pmod{m}$.

Because $tm \equiv 0 \pmod{m}$, it follows that $sa \equiv 1 \pmod{m}$.

Consequently, s is an inverse of a modulo m .

That this inverse is unique modulo m is left as exercise.

Find an inverse of 3 modulo 7 by first finding Bézout coefficients of 3 and 7.

Because $\gcd(3, 7) = 1$, the theorem tells us that an inverse of 3 modulo 7 exists.

The Euclidean algorithm is $7 = 2 \cdot 3 + 1$.

From this equation we see that $(-2) \cdot 3 + 1 \cdot 7 = 1$.

So -2 is an inverse of 3 modulo 7.

Find an inverse of 101 modulo 4620.

Answer Q1

Solving linear congruences

Solve $3x \equiv 4 \pmod{7}$.

Solution: We know that -2 is an inverse of 3 modulo 7 .

Multiplying by -2 we obtain $-2 \cdot 3x \equiv -8 \pmod{7}$.

Hence, $x \equiv -8 \pmod{7}$.

$x = \dots, -8, -1, 6, 13, 20, 27, 34, \dots$

Answer Q2

System of congruences

In the first century, the Chinese mathematician Sun-Tsu asked:

There are certain things whose number is unknown. When divided by 3, the remainder is 2; when divided by 5, the remainder is 3; and when divided by 7, the remainder is 2. What will be the number of things?

This puzzle can be translated into the following question: What are the solutions of the systems of congruences

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}?$$

Solution: $x = 3t + 2 \implies 3t + 2 \equiv 3 \pmod{5} \implies 3t \equiv 1 \pmod{5}$.

Since 2 is an inverse of 3 modulo 5, $t \equiv 2 \pmod{5}$. So $t = 5u + 2$ and so $x = 15u + 8$.

$$15u + 8 \equiv 2 \pmod{7} \implies 15u \equiv -6 \pmod{7} \implies 15u \equiv 1 \pmod{7}.$$

Since 1 is an inverse of 15 modulo 7, $u \equiv 1 \pmod{7}$. So $u = 7v + 1$ and so $x = 105v + 23$ and so $x \equiv 23 \pmod{105}$.

The Chinese Remainder Theorem

Theorem

Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers greater than one and $a_1, a_2, \dots, a_n$ arbitrary integers. Then the system

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$\vdots$

$$x \equiv a_n \pmod{m_n}$$

has a unique solution modulo $m = m_1 m_2 \cdots m_n$.

The solution in the proof of the theorem is constructed as follows:

Let $M_k = m/m_k$, y_k is an inverse of M_k modulo m_k .

Then $x = a_1 M_1 y_1 + a_2 M_2 y_2 + \dots + a_n M_n y_n \pmod{m}$.

Solution: Let $m = 3 \cdot 5 \cdot 7 = 105$, $M_1 = m/3 = 35$, $M_2 = m/5 = 21$, and $M_3 = m/7 = 15$. $y_1 = 2$ is an inverse of $M_1 = 35$ modulo 3, $y_2 = 1$ is an inverse of $M_2 = 21$ modulo 5, $y_3 = 1$ is an inverse of $M_3 = 15$ modulo 7.

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 233 = 23 \pmod{105}.$$

Computer Arithmetic with Large Integers

Suppose that performing arithmetic with integers less than 100 on a certain processor is much quicker than doing arithmetic with larger integers.

For example, we can use the moduli of 99, 98, 97, and 95.

By the Chinese remainder theorem, every non-negative integer less than $99 \cdot 98 \cdot 97 \cdot 95 = 89,403,930$ can be represented uniquely by its remainders when divided by these four moduli.

We represent 123,684 as $(33, 8, 9, 89)$, 413,456 as $(32, 92, 42, 16)$.

We find $123,684 + 413,456 = (33, 8, 9, 89) + (32, 92, 42, 16) =$
 $(65 \bmod 99, 100 \bmod 98, 51 \bmod 97, 105 \bmod 95) = (65, 2, 51, 10)$.

Next solve the system of congruences

$$x \equiv 65 \pmod{99}$$

$$x \equiv 2 \pmod{98}$$

$$x \equiv 51 \pmod{97}$$

$$x \equiv 10 \pmod{95}.$$

Fermat's Little Theorem

Theorem

If p is prime and a is an integer not divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$. Furthermore, for every integer a we have $a^p \equiv a \pmod{p}$.

Fermat's little theorem is extremely useful in computing the remainders modulo p of large powers of integers.

Find $7^{222} \bmod 11$.

Solution: By Fermat's little theorem we know that $7^{10} \equiv 1 \pmod{11}$. So $(7^{10})^k \equiv 1 \pmod{11}$ for every positive integer k . Divide the exponent 222 by 10, finding that $222 = 22 \cdot 10 + 2$. We see that $7^{222} = 7^{22 \cdot 10 + 2} = (7^{10})^{22} 7^2 = (1)^{22} \cdot 49 \equiv 5 \pmod{11}$.

Note that Fermat's Little Theorem is not applicable to find $17^{55} \bmod 111$ and $14^{222} \bmod 7$.

Answer Q3

Homework:

- The paragraphs “Pseudoprimes” at p. 281 and “Primitive Roots and Discrete Logarithms” at p. 283 are left for independent study.
- Solve all odd exercises at p. 284.

Applications of Congruences

(Paragraph 4.5)

Hashing functions

- The central computer at an insurance company maintains records for each of its customers.
- How can memory locations be assigned so that customer records can be retrieved **quickly**?
- The solution to this problem is to use a suitably chosen **hashing function**.
- For instance, customer records are often identified using the **Social Security number** of the customer as the **key**.
- In practice, many different hashing functions are used.
- One of the most common is the function $h(k) = k \bmod m$, where m is the number of available memory locations.

Find the memory locations assigned by the hashing function $h(k) = k \bmod 111$ to the records of customers with Social Security numbers 064212848 and 037149212.

Solution: 064212848 is assigned to memory location 14, because

$$h(064212848) = 064212848 \bmod 111 = 14.$$

Because $h(037149212) = 037149212 \bmod 111 = 65$, 037149212 is assigned to 65.

Because a hashing function is not one-to-one, more than one file may be assigned to a memory location.

When this happens, we say that a **collision occurs**.

Possible resolution of a collision

After making the assignments of records to memory locations in Example on previous slide, assign a memory location to the record of the customer with Social Security number 107405723.

Solution: First note that the hashing function $h(k) = k \bmod 111$ maps the Social Security number 107405723 to location 14, because
 $h(107405723) = 107405723 \bmod 111 = 14$.

However, this location is already occupied (by the file of the customer with number 064212848).

But, because memory location 15, the first location following memory location 14, is free, we assign the record of the customer with Social Security number 107405723 to this location.

Pseudorandom Numbers

- Randomly chosen numbers are often needed for computer simulations.
- Different methods have been devised for generating numbers that have properties of randomly chosen numbers.
- Because numbers generated by systematic methods are not truly random, they are called **pseudorandom numbers**.
- The most commonly used procedure for generating pseudorandom numbers is the **linear congruential method**.

We choose four integers:

the modulus m ,

multiplier a , with $2 \leq a < m$

increment c , with $0 \leq c < m$

seed x_0 , with $0 \leq x_0 < m$.

We generate a sequence of pseudorandom numbers $\{x_n\}$, with $0 \leq x_n < m$ for all n , by successively using the recursively defined function

$$x_{n+1} = (ax_n + c) \bmod m.$$

Example

$$x_{n+1} = (ax_n + c) \bmod m.$$

For certain choices of m , a , c , x_0 , the sequence $\{x_n\}$ becomes periodic. That is, after a certain point, the sequence begins to repeat.

Low periods lead to poor generators.

For these reasons, very large numbers are used in practice.

Let $m = 17$, $a = 5$, $c = 2$, $x_0 = 3$. Then the sequence is as follows.

$$x_1 = (5 \cdot x_0 + 2) \bmod 17 = 0$$

$$x_2 = (5 \cdot x_1 + 2) \bmod 17 = 2$$

$$x_3 = (5 \cdot x_2 + 2) \bmod 17 = 12$$

$$x_4 = (5 \cdot x_3 + 2) \bmod 17 = 11$$

$$x_5 = (5 \cdot x_4 + 2) \bmod 17 = 6$$

$$x_6 = (5 \cdot x_5 + 2) \bmod 17 = 15$$

$$x_7 = (5 \cdot x_6 + 2) \bmod 17 = 9$$

$$x_8 = (5 \cdot x_7 + 2) \bmod 17 = 13 \text{ etc.}$$

Homework:

- The paragraphs “Check digits” at p.; 289
- Solve all odd exercises at p. 292.

Cryptography

(Paragraph 4.5)

Caesar cipher

Cryptography is the study of secure communication via encryption.

One of the earliest uses was in ancient Rome and involved what is now known as a **Caesar cipher**.

This simple encryption system involves a **shift of letters** in a fixed alphabet.

Encryption and decryption is simple modular arithmetic.

In general, we fix an **alphabet**, Σ and let $m = |\Sigma|$.

Second, we fix an **secret key**, an integer k such that $0 < k < m$.

Then the encryption and decryption functions are

$$e_k(x) = (x + k) \bmod m$$

$$d_k(y) = (y - k) \bmod m.$$

Cryptographic functions must be one-to-one (why?).

Example

Let $\Sigma = \{A, B, C, \dots, Z\}$ so $m = 26$. Choose $k = 7$. Encrypt “HANK” and decrypt “KLHU”.

A	B	C	D	E	F	G	H	I	J	K	L	M
↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
13	14	15	16	17	18	19	20	21	22	23	24	25

“HANK” can be encoded (7-0-13-10),

so

$$e(7) = (7 + 7) \bmod 26 = 14$$

$$e(0) = (0 + 7) \bmod 26 = 7$$

$$e(13) = (13 + 7) \bmod 26 = 20$$

$$e(10) = (10 + 7) \bmod 26 = 17$$

so the encrypted word is “OHUR”.

“KLHU” is encoded as (10-11-7-20),

so

$$e(10) = (10 - 7) \bmod 26 = 3$$

$$e(11) = (11 - 7) \bmod 26 = 4$$

$$e(7) = (7 - 7) \bmod 26 = 0$$

$$e(20) = (20 - 7) \bmod 26 = 13$$

So the decrypted word is “DEAN”.

Affine cipher

Clearly, the Caesar cipher is insecure-the key space is only as large as the alphabet. An alternative (though still not secure) is what is known as an **affine cipher**.

Here the encryption and decryption functions are as follows.

$$e(x) = (ax + b) \bmod m$$

$$d(y) = \bar{a}(y - b) \bmod m$$

The function $(ax + b) \bmod m$ is a bijection if and only if $\gcd(a, m) = 1$.

Example

Let $a = 3$, $b = 7$. Encrypt the word “PROOF” and decrypt “UTTQ CTOA”.

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
13	14	15	16	17	18	19	20	21	22	23	24	25

“PROOF” can be encoded as (16-18-15-15-6).

The encryption is as follows.

$$e(16) = (3 \cdot 16 + 7) \bmod 26 = 3$$

$$e(18) = (3 \cdot 18 + 7) \bmod 26 = 9$$

$$e(15) = (3 \cdot 15 + 7) \bmod 26 = 0$$

$$e(15) = (3 \cdot 16 + 7) \bmod 26 = 0$$

$$e(6) = (3 \cdot 6 + 7) \bmod 26 = 25$$

The encrypted message is “DJAAZ”.

$$\bar{3} = 9 \text{ so } d(y) = 9(y - 7) \bmod 26$$

“UTTQ CTOA” can be encoded as (20-19-19-16 2-19-14-0). The decryption is.

$$d(20) = (9 \cdot 13) \bmod 26 = 13$$

$$e(19) = (9 \cdot 12) \bmod 26 = 4$$

$$e(16) = (9 \cdot 9) \bmod 26 = 3$$

$$e(2) = (9 \cdot (-5)) \bmod 26 = 7$$

$$e(14) = (9 \cdot 7) \bmod 26 = 11$$

$$e(0) = (9 \cdot (-7)) \bmod 26 = 15$$

The encrypted message is “NEED HELP”.

Answer Q4

Public Key Cryptography

- All classical ciphers, including shift and affine ciphers, are **private** key cryptosystems.
- Knowing the encryption key allows one to quickly determine the decryption key.
- All parties who wish to communicate using a private key cryptosystem must share the key and keep it a secret.
- In **public** key cryptosystems, first invented in the 1970s, knowing how to encrypt a message does not help one to decrypt the message.
- Therefore, everyone can have a publicly known encryption key.
- The only key that needs to be kept secret is the decryption key.

The RSA Cryptosystem

- A public key cryptosystem, now known as the RSA system was introduced in 1976 by three researchers at MIT.
- It is now known that the method was discovered earlier by Clifford Cocks, working secretly for the UK government.
- The public encryption key is a pair (m, a) where the modulus m is the product of two large (200 digits) primes p and q and exponent a is relatively prime to $(p - 1)(q - 1)$.
- Factorization $m = p \cdot q$ is kept private! With approximately 400 digits, m cannot be factored in a reasonable length of time.

The RSA system works as follows.

- Choose 2 (large) primes p, q .
- Compute $m = p \cdot q$.
- Compute $\phi(m) = (p - 1)(q - 1)$.
- Choose a , $2 \leq a \leq \phi(m)$ such that $\gcd(a, \phi(m)) = 1$.
- Publish m, a .
- Keep p, q private.

Then the encryption function is

$$e(x) = x^a \bmod m.$$

Example

Encrypt the message “STOP” using the RSA cryptosystem with key $(2537, 13)$.

Solution: To encrypt, we first translate the letters in “STOP” into their numerical equivalents 18191415.

We then group these numbers into blocks of four digits (because $2525 < 2537 < 252525$), to obtain 1819 1415.

We encrypt each block using the mapping $f(x) = x^{13} \bmod 2537$.

Computations using modular exponentiation algorithm show that

$$1819^{13} \bmod 2537 = 2081$$

$$1415^{13} \bmod 2537 = 2182$$

The encrypted message is 2081 2182.

- For a given p, q , we first compute $b = \text{inverse of } a \text{ modulo } \phi(m) = (p - 1)(q - 1)$.
- Remember that we choose a to be relatively prime with $\phi(m)$.
- The decryption function is

$$d(x) = x^b \bmod m.$$

Example

We receive the encrypted message 0981 0461. What is the decrypted message if it was encrypted using the RSA cryptosystem with key $(2537, 13)$, with $p = 43$ and $q = 59$.

Note that $2537 = 43 \cdot 59$, $p = 43$ and $q = 59$ are primes, and $\gcd(a, (p-1)(q-1)) = \gcd(13, 42 \cdot 58) = 1$.

Solution: $b = 937$ is an inverse of 13 modulo $42 \cdot 58 = 2436$.

The decryption function is

$$d(x) = x^{937} \bmod 2537.$$

we use the modular exponentiation algorithm to compute

$$0981^{937} \bmod 2537 = 0704$$

$$0461^{937} \bmod 2537 = 1115.$$

Consequently, the numerical version of the original message is 07041115.

Translating back to English, the message is "HELP".

Homework:

- Read “Cryptographic Protocol” at p. 302
- Solve *odd* exercises starting at p. 304.

Answer Q5